

Social Media

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Social media platforms for research



1 Introduction to Social Media Data

- Benefits & Limitations of Social Media Data

2 Types of Social Media Data and APIs

- Reddit
- Twitter/X
- Facebook & Instagram
- TikTok

3 Summary

What is Social Media Data?

- Data generated by users on social media platforms
- Includes text, images, videos, links, and metadata
- Can be used to study a wide range of topics
- Can be collected using APIs, web scraping, or manual data collection
- Can be analyzed using a wide range of methods

Why Study Social Media Data?

- Social media data are **ubiquitous**
- Social media data are **rich**
- Social media data are **real-time**
- Social media data are **interactive**
- Social media data are **cost-effective**
- Social media data can be used to study **sensitive topics**

Forms of Social Media Data

- **Text**: posts, comments, messages, etc.
- **Images**: photos, memes, etc.
- **Videos**: TikToks, YouTube videos, etc.
- **Links**: URLs, hyperlinks, etc.
- **Metadata**: timestamps, geotags, etc.
- **Networks**: connections between users, groups, etc.
- **Engagement**: likes, shares, comments, etc.

Social Media Data Collection Methods

- **APIs**: Application Programming Interfaces
 - **Streaming APIs**: real-time data collection
 - **REST APIs**: historical data collection
 - Example: Reddit API
- **Web scraping**: automated data collection from websites
- **Manual data collection**: copying and pasting data
- **Hybrid methods**: combining methods
- **Data sharing**: using data collected by others

Benefits of Social Media Data

- **Large-scale:** millions of users, billions of posts
- **Real-time:** data are available as events unfold
- **Rich:** text, images, videos, links, and metadata
- **Interactive:** users can respond to each other
- **Cost-effective:** data are often free or low-cost
- **Ethical:** can be used to study sensitive topics

Limitations of Social Media Data

- **Accessibility** e.g., private posts are protected; access/rules can change any time
- **Non-representativeness**: users are non-random sample of the general population; what users share is non-random
- **Reproducibility** e.g., companies may limit what researchers can make public
- **Rules and terms of service**: written to be (intentionally) ambiguous
- **Ethics**: considerations include how the data are used, (in)ability to gain consent from research subjects, and protecting research subjects from harm

- Multiple forms of research:
 - **Platform focused**: studying the platform itself
 - **As a data source**: using data to answer research questions

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3 Summary

- Forum for discussion, news aggregation, and content rating
- Subreddits (communities) are both user-created and user-moderated
- Used for research across disciplines, but (so far) rarely political science
 - Some examples include:
 - political subreddits (e.g., r/politics, r/worldnews, and r/The_Donald)
 - mental health subreddits (e.g., r/depression, r/suicidewatch, r/anxiety, r/bipolarreddit)
 - drug use subreddits (e.g., r/opiates)

Benefits

- 100 million daily active users worldwide
- Accessible via APIs at no cost
- Can be used to target people with specific interests
- Can be used to study sensitive topics
- Can be used to study niche communities

Challenges

- **Representativeness**: skews young, male and conservative
- **Pseudonyms** are the norm; user location information often unavailable
- Posting is **strategic**; top posts get more interactions
- Additional **ethical considerations** may apply to certain populations
- Subreddits have **own norms, cultures, and moderation practices**
- **Rules can change** at any time

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Article | [Open access](#) | Published: 02 February 2021

No echo in the chambers of political interactions on Reddit

[Gianmarco De Francisci Morales](#) ✉, [Corrado Monti](#) ✉ & [Michele Starnini](#) ✉

[Scientific Reports](#) **11**, Article number: 2818 (2021) | [Cite this article](#)

34k Accesses | **89** Altmetric | [Metrics](#)

- Are voters polarized?
- Reddit users who are Clinton supporters shown to be likely to respond to out-group partisans
- Contradicts previous findings of homophilous networks
- **Methodology**: collected data from 2016 election-related subreddits
- Contradicts previous research on echo chambers and filter bubbles

Benefits

- 100 million daily active users worldwide
- Mostly public (if you're rich)
- Can be used to target people with specific interests
- Used for research across disciplines, including political science
- Over-represents elites

Challenges

- **Representativeness**: skews young
- **Pseudonyms** are the norm; user location information often unavailable
- Posting is **strategic**; top posts get more interactions
- **Rules can change** at any time
- **Restricted accessibility** (varies by API type)

Documentation

Search the docs



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[API reference index](#)

X API



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[X API v2](#)



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- Access to Ads API
- \$200 per month
- \$175 per month with annual subscription (\$2,100 total)
- 2 Top-ups allowed

Pro

For startups scaling their business

- Rate-limited access to suite of v2 endpoints, including search and filtered stream
- 1,000,000 Posts per month - GET at the app level
- 300,000 Posts per month - posting limit at the app level
- 1 Project
- 3 Apps per Project with unique Environment (Development/ Production/ Staging)
- Login with X
- Access to Ads API
- \$5,000 per month
- \$4,500 per month with annual subscription (\$54,000 total)
- 2 Top-ups allowed

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OPEN ACCESS Check for updates

The Effects of COVID-19 Infection on Opposition to COVID-19 Policies: Evidence from the U.S. Congress

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ABSTRACT

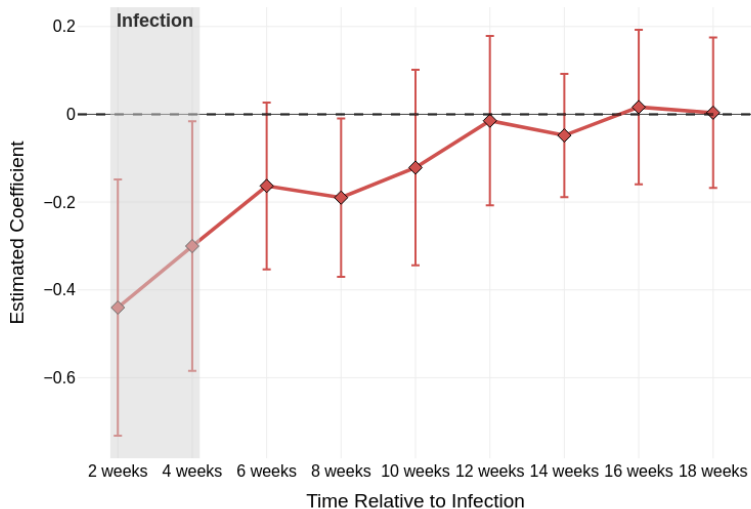
Elites' skepticism of scientific consensus presents a formidable challenge in addressing critical issues like climate change and global pandemics. While extensive research has explored the capacity of events related to these challenges to act as "exogenous shocks," motivating the general public to reassess their risk perceptions, our understanding of how elites similarly respond to such shocks remains limited. In this article, we investigate whether COVID-19 infections influenced US lawmakers' support for COVID-19 containment measures, focusing on expressed opposition to COVID-19 policies on social media and in press releases throughout the first two years of the pandemic. Employing a staggered difference-in-differences design and matrix completion methods, our analysis reveals that COVID-19 infections caused a reduction of approximately 30% in legislators' expressions of opposition to COVID-19 policies on social media. These findings underscore that elites are indeed responsive to policy shocks – even in highly polarized contexts – when they are personally affected by an issue.

KEYWORDS

COVID-19; polarization; partisanship; political communication; political behavior

- Do US congress members change their positions on COVID-19 after being infected?
- **Methodology**: collected all tweets by congress members mentioning COVID-19
- Classified tweets as **pro-** or **anti** COVID-19 policy
- Used **staggered difference-in-differences** to estimate the effect of being infected on policy positions
- Found that being infected with COVID-19 **reduced** the likelihood of opposing COVID-19 policies on Twitter (but only for a bit)

Dickson and Yildirim (2024)



Matrix Completion Estimates and 95% confidence intervals

Polit Behav (2017) 39:629–649
DOI 10.1007/s11109-016-9373-5



ORIGINAL PAPER

Tweetment Effects on the Tweeted: Experimentally Reducing Racist Harassment

Kevin Munger¹

Published online: 11 November 2016
© Springer Science+Business Media New York 2016

Abstract I conduct an experiment which examines the impact of group norm promotion and social sanctioning on racist online harassment. Racist online harassment de-mobilizes the minorities it targets, and the open, unopposed expression of racism in a public forum can legitimize racist viewpoints and prime ethnocentrism. I employ an intervention designed to reduce the use of anti-black racist slurs by white men on Twitter. I collect a sample of Twitter users who have harassed other users and use accounts I control (“bots”) to sanction the harassers. By varying the identity of the bots between in-group (white man) and out-group (black man) and by varying the number of Twitter followers each bot has, I find that subjects who were sanctioned by a high-follower white male significantly reduced their use of a racist slur. This paper extends findings from lab experiments to a naturalistic setting using an objective, behavioral outcome measure and a continuous 2-month data collection period. This represents an advance in the study of prejudiced behavior.

Keywords Online harassment · Social media · Randomized field experiment · Social identity

- **Research question:** Can a single tweet reduce racist harassment on Twitter?
- **Methodology:** randomly assigned users to receive a message opposing racist language
- **Results:** In-group norm promotion is influential in reducing racist language on social media

Benefits

- 3.35 billion daily active users worldwide (across all Meta apps)
- Somewhat public
- Offers other forms of curated data (e.g., ads)
- Can be used to target people with specific interests
- Used for research across disciplines, including political science

Challenges

- Meta **tightly controls** the information researchers can use from its platforms (bans scraping and has taken action against violators of TOS)
- **Representativeness**: skews young[?]
- **Rules can change** at any time

- Meta Content Library and API
- Meta Data for Good

Meta Data for Good

Infrastructure

Electrical Distribution Grid Maps

This predictive model can map medium-voltage electrical distribution infrastructure in any country.

[→ Learn More](#)

Inclusive Internet Index

The Inclusive Internet Index measures whether the internet is accessible and affordable, and how it affects social and economic mobility.

[→ Learn More](#)

Commuting Zones

Commuting Zones are geographic areas where people live and work and are useful for understanding local economies, as well as how they differ from traditional boundaries.

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Infrastructure Route Study

This Infrastructure Route Study contains a collection of open source data collected for expanding fiber routes to extend Internet connectivity in the Democratic Republic of the Congo.

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Global Urban Areas

The Global Urban Areas dataset is a high resolution dataset consists of 37,000 polygonal areas around the globe, and provides enough detail for most use-cases.

[→ Learn More](#)

Network Coverage Maps

Network Coverage Maps show where people on Facebook have cellular connectivity.

[→ Learn More](#)

Base

The Base Dataset is a foundational element of navigation, local search, and many other location-based applications.

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Buildings

An alpha version of open map data showcasing over 6 million buildings that have been derived from the US Geological Survey (USGS) 3D Elevation Program Light Detection and Ranging and open building footprint data.

[→ Learn More](#)

Divisions

The Overture Maps Foundation (OMF) divisions dataset provides information on official and non-official organizations of people at the country, region, province, city, and neighborhood level.

[→ Learn More](#)

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Elite Cues and Noncompliance

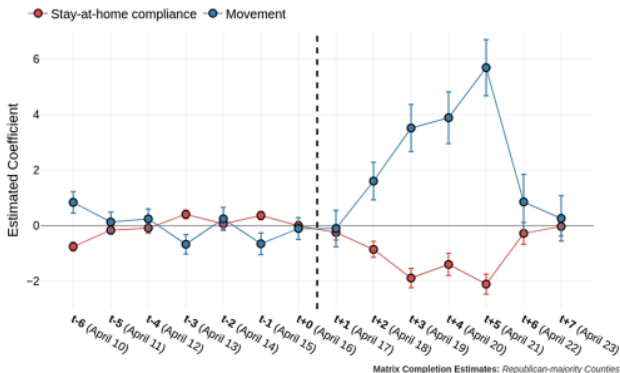
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Political leaders increasingly use social media to speak directly to voters, but the extent to which elite cues shape offline political behavior remains unclear. In this article, we study the effects of elite cues on noncompliant behavior, focusing on a series of controversial tweets sent by US President Donald Trump calling for the “liberation” of Minnesota, Virginia, and Michigan from state and local government COVID-19 restrictions. Leveraging the fact that Trump’s messages exclusively referred to three specific US states, we adopt a generalized difference-in-differences design relying on spatial variation to identify the causal effects of the targeted cues. Our analysis shows that the President’s messages led to an increase in movement, a decrease in adherence to stay-at-home restrictions, and an increase in arrests of white Americans for crimes related to civil disobedience and rebellion. These findings demonstrate the consequences of elite cues in polarized environments.

- **Research question:** Can elites mobilize supporters via social media?
- **Methodology:** Use Meta's Mobility During Crisis data to estimate the effects of Trump's messages on compliance with COVID restrictions
- **Results:** Trump's messages decreased compliance with COVID restrictions in areas where he was popular

FIGURE 5. Dynamic Effects of “Liberate” Cues on Mobility in Republican Counties



Note: Matrix completion coefficient estimates and 95% confidence intervals for the effect of the cues on movement (blue) and stay-at-home compliance (red) in Republican-majority counties (e.g., model 3 in Tables 1 and 2). The counterfactual includes Republican-majority counties around the country that were not targeted in the President calls for liberation and were under the same mandatory state restrictions.

Facebook Research Example:



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HOME > SCIENCE > VOL. 381, NO. 6656 > HOW DO SOCIAL MEDIA FEED ALGORITHMS AFFECT ATTITUDES AND BEHAVIOR IN AN ELECTION CAMPAIGN?

 | SPECIAL ISSUE RESEARCH ARTICLE | SOCIAL MEDIA



How do social media feed algorithms affect attitudes and behavior in an election campaign?

ANDREW M. GUESS , NEIL MALHOTRA , JENNIFER PAN , PABLO BARBERÁ , HUNT ALLCOTT, TAYLOR BROWN , ADRIANA CRESPO-TENORIO, DREW DIMMERY 
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 35,225  103



- Does the Facebook algorithm expose people to misinformation and hostile content?
- **Methodology**: Working with Facebook, new feeds were randomly switched to chronological order
- **Results**: Engagement decreased when feeds were switched to chronological order
- **Results**: Altering the new feed algo did **not** significantly alter levels of issue polarization, affective polarization, and political knowledge (with some exceptions)

Images, Politicians, and Social Media: Patterns and Effects of Politicians' Image-Based Political Communication Strategies on Social Media

The International Journal of Press/Politics
2021, Vol. 26(1) 119–142
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Xénia Farkas¹  and Márton Bene² 

Abstract

Although images have always been part of politics, research on the visual aspects of political communication recently gained momentum, especially with the spread of social media-based political communication. However, there are still several significant research gaps in this field. The aim of this article is to identify and compare the patterns and effects of Hungarian politicians' ($n = 51$) image-based communication on Facebook ($n = 2,992$) and Instagram ($n = 868$) during the Hungarian parliamentary election campaign in 2018. By doing so, we shed light on two important dimensions of personalization: individualization and privatization. This work is designed to fill three gaps in the literature. We argue that existing research of visual political communication (1) treats images predominantly as illustrations, (2) is limited to single-platform studies, and (3) does not investigate the engagement effects of images. To move beyond these limitations, this study investigates images as objects of interest on their own; it adopts a cross-platform comparative approach and examines the engagement effects of visual cues by applying a combination of inductive and deductive qualitative content analysis. Our results show that images are often used to personalize communication. While on Facebook the individualization dimension of personalization is more common and popular, on Instagram its privatization dimension prevails. Furthermore, on Facebook, users like more politics-related candidate-centered images, but on Instagram we could not find similar effects for more informal visuals.

- **Research question:** How do politicians construct their images on social media platforms
- **Data:** Compare Hungarian politicians' images on Facebook and Instagram
- **Results:** Politicians use Facebook to individualize
- **Results:** Politicians use Instagram to privatize

- Over 1 billion monthly active users
- **Controversial**: Concern about user data privacy / national security
 - but note that Facebook collects even more data about you!
- **Young** user base
- Some API access for researchers in the US and Europe

TikTok Research Example

Fifteen Seconds of Fame: TikTok and the Supply Side of Social Video

Benjamin Guinaudeau
University of Konstanz

Kevin Munger
Pennsylvania State University

Fabio Votta
University of Amsterdam

Abstract

TikTok has rapidly developed from a punchline for jokes about “kids these days” into a formidable force in American politics. The speed of this development is unprecedented, even in the rapidly-changing world of digital politics. Through a combination of hashtag and snowball sampling, we identify 11,546 TikTok accounts who primarily post about politics, allowing us to analyze trends in the posting, viewing and commenting behavior on 1,998,642 tiktoks they have uploaded. We test a number of theories about how the unique combination of affordances on TikTok shapes how it is used for political communication. Compared to the dominant platform for political videos (YouTube) we find that a higher percentage of TikTok users upload videos, TikTok view counts are more dominated by virality, and viewership of videos are less dependent on a given accounts’ number of followers/subscribers. We discuss how these findings affect the production of content that ultimately determines the experience of TikTok consumers.

- **Research question:** How does TikTok differ from YouTube?
- **Data:** Identify 11,546 TikTok accounts who primarily post about politics
- **Results:** A higher percentage of TikTok users upload videos
- **Results:** TikTok view counts are more dominated by virality
- Viewership of videos are less dependent on a given accounts' number of followers/subscribers.

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- TikTok

3 Summary

Summary

- Social media data are **ubiquitous**
- Social media data are **rich**
- Social media research can be **cost-effective**
- Different approaches with [arguably] different strengths and weaknesses
- What do we learn from social media research and how might it generalize?